

A Sequence-to-Sequence Model Based on Attention Mechanism for Wave Spectrum Prediction

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Abstract—Ocean wave spectrum are mainly estimated from wave surface records, and traditional methods involve physically-based modeling of wave dynamics and frequency analysis. In this paper, we employ machine learning strategy and propose a Seq2Seq (sequence to sequence) model which connects the encoder and decoder with an attention mechanism. This model can both effectively predict wave spectrum and be easily implemented. Experiments on numerical simulations show the feasibility of the proposed model in wave spectrum estimation and the accuracy comparing with traditional methods.

Index Terms—wave spectrum, Seq2Seq, attention

I. INTRODUCTION

Wave spectrum describes the internal energy distribution of ocean waves over frequencies, which is also known as power spectral density (PSD). It has a very wide range of applications [1]. Spectrum can be used to infer wave height, period and other elements for wave forecasting. In ocean engineering, the spectrum can describe the complex structure of perplexed waves. In marine environment, the complex and non-stop exchange process of momentum, energy and matter between ocean waves and the atmosphere is closely related, and it also affects the magnetic field changes on the sea surface and the propagation of sound waves. The wave spectrum is very important for studying the environmental conditions inside and outside the ocean. Better estimation of wave spectrum is of great significance in both academia and engineering.

The wave spectrum cannot be directly observed, and can only be estimated from measurement of specific devices, such as wave buoy. Therefore, estimation of wave spectrum has been one of the fundamental topics in ocean wave research.

Existing methods of wave spectrum estimation is theoretically and empirically based, and it was found to well fit the observation data. Although extensive research has been carried out on wave spectrum estimation and a large number of spectrum types have been proposed [2]–[4], limitations still exist, such as the subjective selection of spectrum forms, relevant parameters, observation and verification conditions. In addition, these methods usually require large amount of calculations in the solution process.

This paper aims to introduce a deep learning method to address these problems through end-to-end learning. The merit of the deep learning method is that it can be quickly modeled through learning to avoid solving complex differential equations. A Seq2Seq model based on the attention mechanism [5]

is proposed, which only needs to input the acceleration data obtained by the sensor as a time series and the spectrum will be predicted. The contributions of this work are summarized as follows:

- We innovatively consider wave spectrum estimation as a sequence-to-sequence prediction problem, and propose an end-to-end deep neural network with attention mechanism.
- We conducted extensive experiments to prove our approach can achieve better results than traditional methods on our simulation dataset.

II. RELATED WORK

A. Physically-based methods

For half a century, estimation of the power spectral density (PSD) has been extensively studied. In 1952, WJ Pierson [6] proposed the covariance function estimation spectrum and applied it to describe ocean waves. Later on, two spectral estimation techniques based on Fourier transform have been developed. One is based on the direct FFT (Fast Fourier Transform) operation and is referred to as periodogram, which can be traced back to Schuster [7] in 1899. The other one is an indirect approach based on the autocorrelation estimate proposed by Blackman and Tukey in 1958 [8]. The FFT method is computationally efficient and can produce reasonable results for a large number of signal processing occasions. However, the most prominent limitation of the FFT method is the low frequency resolution. In order to alleviate the inherent limitations of FFT, many alternative spectrum estimation methods were proposed. In 1967, JP Burg proposed the maximum entropy method, also known as MEM [9], which provides better resolution and performance. In nowadays, these methods are important spectral estimation methods that are still often used by oceanography scientists.

B. Seq2Seq

The Seq2Seq model was mainly used for machine translation tasks [10], [11]. The key idea is to encode the source signal sequence as a fixed-length vector and used the decoder to generate a target signal sequence. However, one problem with encoder-decoder networks was that as the length of the input sequence increased, their performance would drop rapidly. In order to solve this problem, the attention mechanism came into

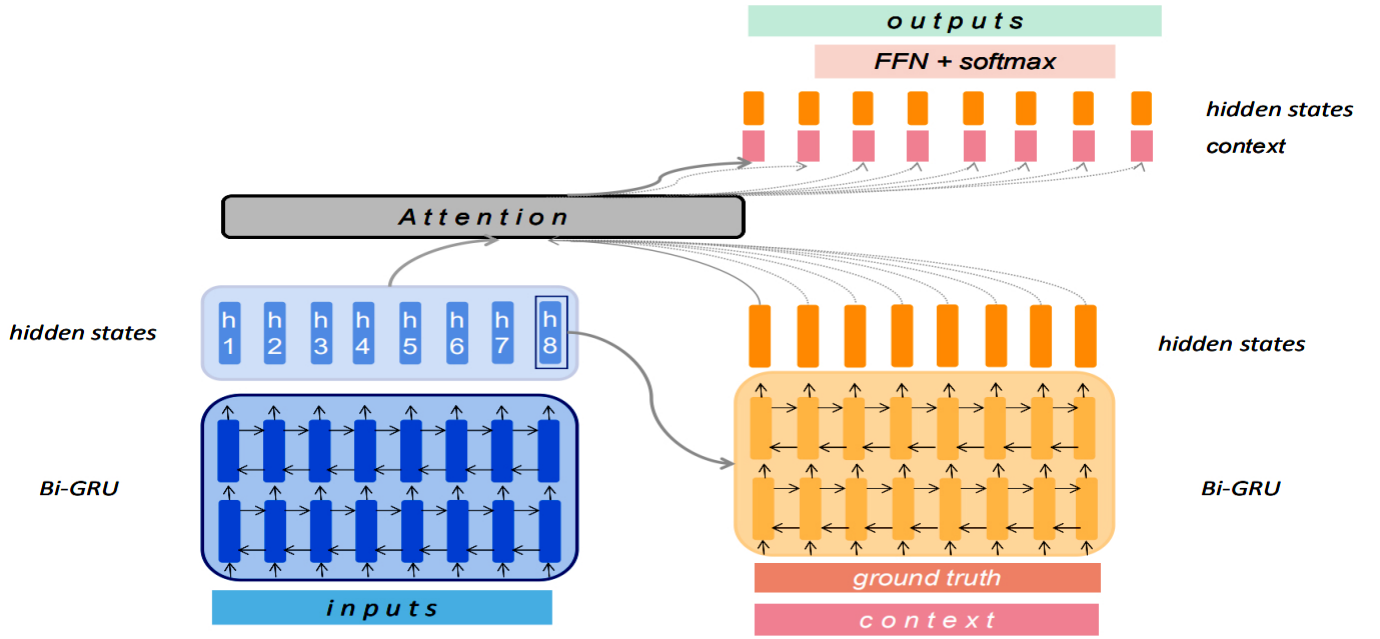


Fig. 1. Graphical illustration of the Seq2Seq model with attention mechanism. The details of the encoder architecture is on the left and the decoder is on the right.

being. In 2014, the attention mechanism was first used to solve the NLP problem [12]. Attention can grasp the important point, even if the text is relatively long, it will not lose important information.

III. MODEL

A time series is defined as a series of actual observations with consecutive time stamps. The acceleration data obtained by the sensor in a continuous period of time is taken as a time series $(x_1, x_2, x_3, \dots, x_n)$, ground truth of target energy density denoted as $\mathbf{Y} = (y_1, y_2, y_3, \dots, y_m)$. Given the input series $(x_1, x_2, x_3, \dots, x_n)$, our goal is to predict the value of the target series $\hat{\mathbf{Y}}$ which will approximate $\mathbf{Y}$ as far as possible. More specifically, the Seq2Seq model aims to learn a nonlinear mapping $F(\cdot)$ such that

$$\hat{\mathbf{Y}} = F(x_1, x_2, x_3, \dots, x_n) \quad (1)$$

Based on above, we use the Seq2Seq model based on the attention mechanism to complete the prediction task. In the encoder, the acceleration sequence is used as input and encoded into a fixed-length vector. In the decoder, a temporal attention mechanism is used to automatically select relevant encoder hidden states across all time steps. A graphical illustration of the proposed model is shown in Fig. 1.

A. Encoder

The encoder is essentially an bidirectional GRU that encodes the input sequences into a feature representation. The bidirectional GRU consists of two GRU cyclic layers with opposite information transfers. The first layer transfers information in chronological order with a forward hidden state

$\vec{h}_i$, and the second layer transfers information in reverse chronological order with a backward hidden state $\overleftarrow{h}_i$. A simple concatenation of two represents the encoder state h_i . This allows it to not only use past information, but also capture future information.

$$h_i = [\vec{h}_i; \overleftarrow{h}_i]^\top, i = 1, 2, 3, \dots, n \quad (2)$$

Compared with LSTM [13], GRU reduces the use of one gate and only keeps the update gate and the reset gate. The simple structure enables it to participate in the calculation more quickly. The GRU illustrated in Fig.2 can be expressed as follows:

$$z_t = \sigma(\mathbf{W}^{(z)}x_t + \mathbf{U}^{(z)}h_{t-1}) \quad (3)$$

$$r_t = \sigma(\mathbf{W}^{(r)}x_t + \mathbf{U}^{(r)}h_{t-1}) \quad (4)$$

$$h'_t = \tanh(\mathbf{W}x_t + r_t \odot \mathbf{U}h_{t-1}) \quad (5)$$

$$h_t = z_t \odot h_{t-1} + (1 - z_t) \odot h'_t \quad (6)$$

where h_{t-1} and h_t are the previous and current hidden state, h_t is final memory at current time step, x_t is the current input, r_t and z_t are the reset gate and update gate. $\mathbf{W}$ and $\mathbf{U}$ are parameters to learn, σ and $\odot$ are a logistic sigmoid function and an elementwise multiplication, respectively.

B. Decoder with attention

To predict the output $\hat{\mathbf{Y}}$, we use another bidirectional GRU to decode the encoded input information. The decoder network has hidden states $s_t = f(s_{t-1}, y_{t-1}, c_t)$ for the output at position $t, t = 1, 2, \dots, m$, where c_t is the context vector.

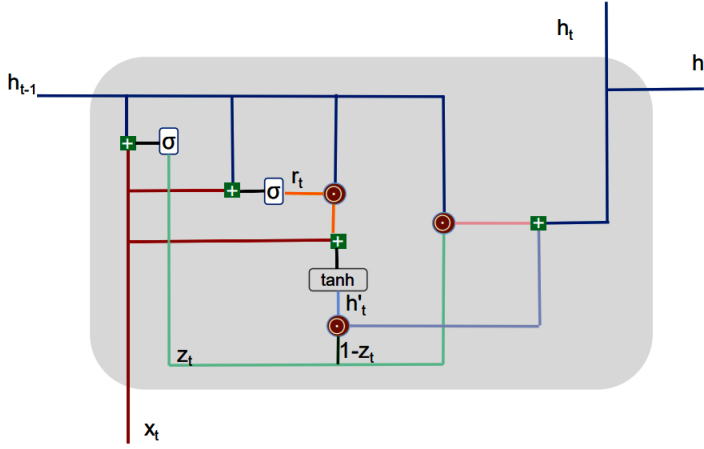


Fig. 2. The simple GRU network.

Since each encoder hidden state h_i is mapped to a temporal component of the input, the attention mechanism computes the context vector c_t as a weighted sum of all the encoder hidden states $h_1, h_2, h_3, \dots, h_t$. Note that the context vector c_t is distinct at each time step.

$$c_t = \sum_{i=1}^T \beta_t^i h_i \quad (7)$$

The attention weight β_t^i represents the importance of the i_{th} encoder hidden state for the prediction.

$$\beta_t^i = \frac{\exp(\text{score}(s_{t-1}, h_i))}{\sum_{i'=1}^T \exp(\text{score}(s_{t-1}, h_{i'}))} \quad (8)$$

The score function is an alignment model used to evaluate the degree of matching between the input and the output information near a certain position. The Alignment model is a feedforward neural network used for scoring, which is jointly trained with the entire model. The calculation is in the following form:

$$\text{score}(s_t, h_i) = V_a^\top \tanh(W_a[s_t; h_i]) \quad (9)$$

Where V and W are weight matrices to be learned. In summary, the hidden states of the decoder are spliced with the context vector c , and the final output is obtained after a series of nonlinear transformations and softmax.

IV. EXPERIMENTS

A. Data preparation

In this experiment, the method of numerical simulation is used to obtain data on acceleration and wave spectral density. The acceleration data is used as the input X of the model, and the wave spectrum density $\hat{Y}$ is predicted through the learning of the network. Therefore, it is necessary to provide the true spectral density as the ground truth. The wave spectral density is not an accurate concept, it can only be estimated by wave surface records. In order to ensure the accuracy of the ground truth, a method of inversely solving the acceleration

data based on the wave spectrum is adopted. We use WAFO [14] to generate the data we need, which is a toolbox of Matlab routines for statistical analysis and simulation of random waves and random loads. Use the Torsethaugen model [15] to get the wave spectrum, which are established from a broad range of combinations of significant wave height $Hm0$ and primary peak period Tp . The range of the measured $Hm0$ and Tp are from 0.5 to 11 meters and from 3.5 to 19 seconds. In the frequency range of 0 to 5.5 rad/s, modify the values of these two parameters to obtain the wave spectral density. Then, use the function `spec2sdat` in WAFO to calculate the displacement in the vertical direction and get corresponding acceleration data after two derivations.

Based on the above, our dataset consists of 2800 pieces of acceleration data with a length of 1024 and the same number of wave spectral density data with a length of 257. Among them, 70% of the data is used as the training set, and 30% of the data is used as the test set.

B. Parameter settings and evaluation metrics

In this model, two layers of bidirectional GRU are used in the encoder and decoder, the number of hidden states on the encoder side is 1024, on the decoder side is 257. The size of the minibatch is 16. We use the Adam optimizer [16] to train the model. In the training process, we adopt the mean squared error as objective function:

$$O(Y, \hat{Y}) = \frac{1}{N} \sum_{i=1}^N (Y_i - \hat{Y}_i)^2 \quad (10)$$

Where N is the number of training samples. The number of iterations for the entire network on the training set is 600.

To evaluate the effectiveness of various methods for wave spectrum prediction, we use two different evaluation metrics to measure the estimated quality. Among them, mean absolute error(MAE) and root mean squared error(RMSE) are scale-dependent measures. MAE is denoted as $MAE = \frac{1}{N} \sum_{i=1}^N |Y_i - \hat{Y}_i|$, RMSE is defined as $RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (Y_i - \hat{Y}_i)^2}$.

TABLE I
THE MAE AND RMSE PERFORMANCE OF DIFFERENT METHODS ON THE WAVE DATASET

Various Models	Wave Dataset	
	MAE	RMSE
Seq2Seq-attention	0.0765	0.3158
COV	0.1847	0.6843
PMEM	0.2117	0.7392
Periodogram	0.1990	0.7120
Periodogram-overlapping	0.0950	0.4383

C. Results

To demonstrate the effectiveness of the Seq2Seq with attention, we compare it against 4 baseline methods. Among them,

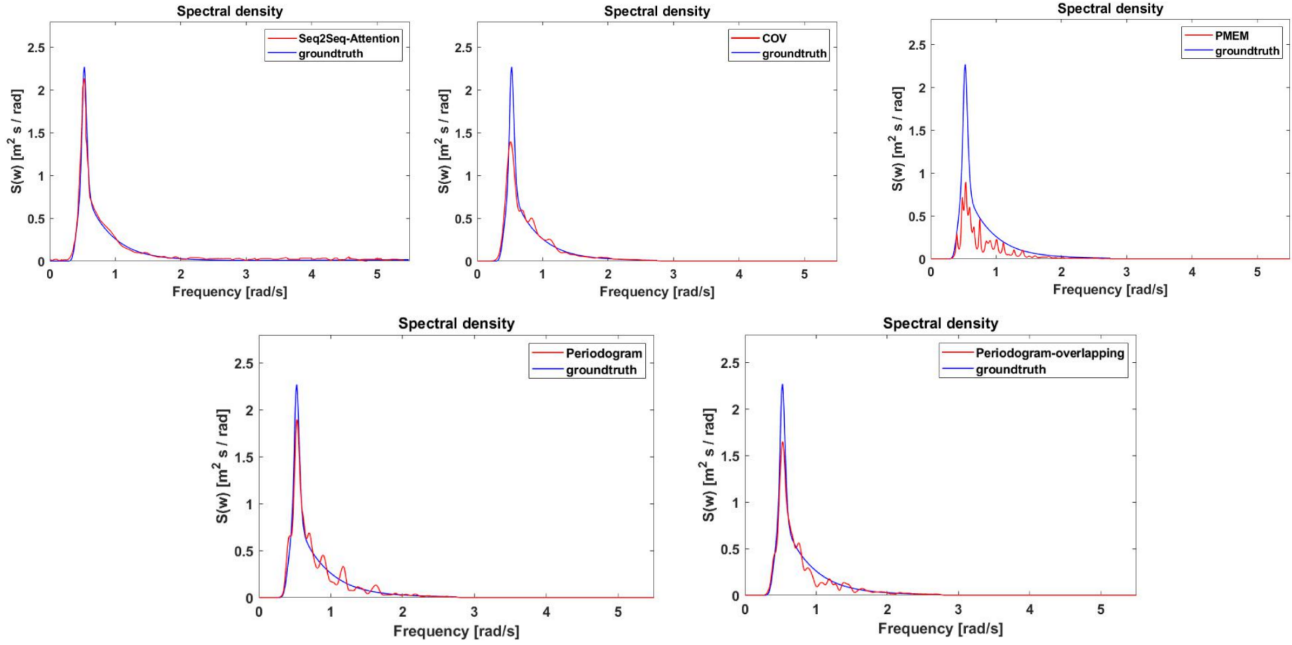


Fig. 3. The prediction results of the method proposed in this paper and the four traditional methods on the single-peak spectrum (the single-peak spectrum indicates that the sea state is dominated by wind).

the COV method uses a parzen window function for frequency smoothing on the estimated autocovariance function, the Periodogram method is the Welch average periodogram method without overlapping batches, the Periodogram-overlapping method is the Welch's averaged periodogram method with overlapping batches, the PMEM method is the maximum entropy method (Periodogram using Yule-Walker AR method [17], [18]).

As is shown in Table I, the experimental results show that the performance of our model on the MAE is 0.0765, and the RMSE is 0.3158. The MAE and RMSE indicate the overall difference between the ground truth and the prediction. It suggests that the Seq2Seq model with attention has better performance.

For visual comparison, we show the prediction result of our model and traditional methods over the same dataset in Fig.3 and Fig. 4. We observe that our model generally fits the ground truth much better than the 4 traditional methods.

V. CONCLUSION

In this paper, a new method was proposed to predict the wave spectrum. We used the Seq2Seq model based on the attention mechanism to verify on the simulated data. In addition, compared with traditional estimation methods, our model obtained better results, which also showed the feasibility of our approach. Moreover, the wave direction spectrum [19] is also a very important concept. It not only contains the second-order

information of the wave, but also directly gives the distribution of wave energy relative to frequency and direction. In the future, we intend to improve our model to predict the wave direction spectrum.

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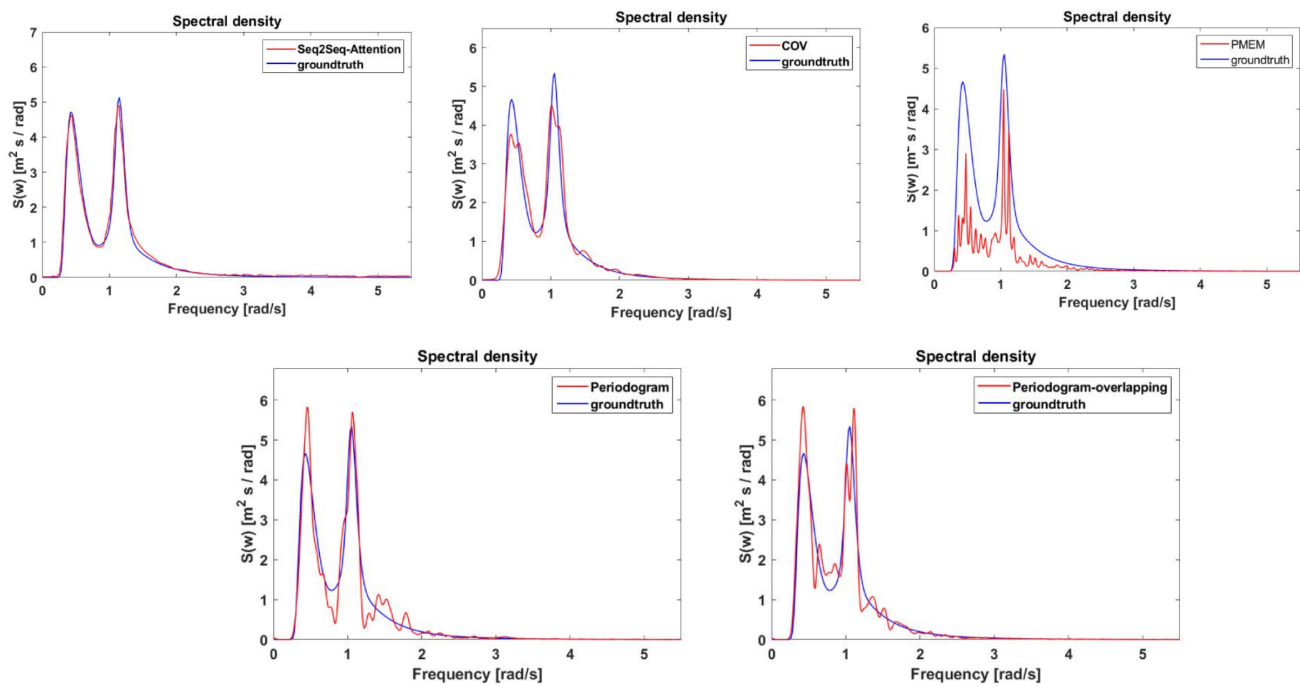


Fig. 4. The prediction results of the method proposed in this paper and the four traditional methods on the double-peak(wind+swell) spectrum.

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